

# Customer Retention & Churn Analysis

## Data Analytics Project for a Subscription-Based Business

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# Project Overview

## Objective

The goal of this project is to analyze customer data from a subscription-based business to identify:

- Key patterns driving customer churn
- Factors influencing customer retention
- Customer usage behavior and engagement
- Support service performance
- Long-term revenue trends and customer lifetime value

## Business Importance

Reducing churn is critical for subscription businesses because recurring revenue and customer lifetime value determine long-term profitability.

Insights from this analysis can help organizations:

- Improve customer retention strategies
- Optimize product features
- Enhance customer support services
- Increase revenue through upgrades and long-term subscriptions

# Dataset Overview

The dataset contains information about:

- Customer accounts
- Subscriptions and plan tiers
- Feature usage data
- System errors
- Support tickets
- Customer churn behavior

## Key Dataset Statistics

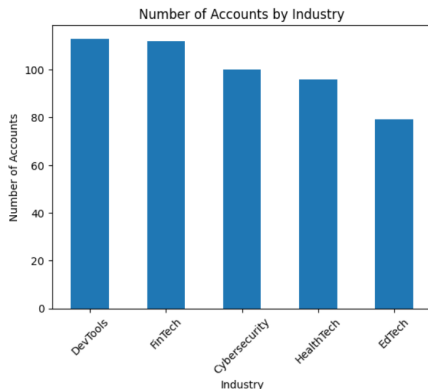
- Total Subscriptions: 5,000
- Customer Churn Count: 600
- Total Monthly Recurring Revenue (MRR): \$11,338,747
- Annual Recurring Revenue (ARR): \$136,064,964
- Average Customer Lifetime Value (CLV): \$10,307.95

# Customer Industry Distribution

The platform serves customers across several technology-focused industries.

| Industry      | Customers |
|---------------|-----------|
| DevTools      | 113       |
| FinTech       | 112       |
| Cybersecurity | 100       |
| HealthTech    | 96        |
| EdTech        | 79        |

# Graph and Insight



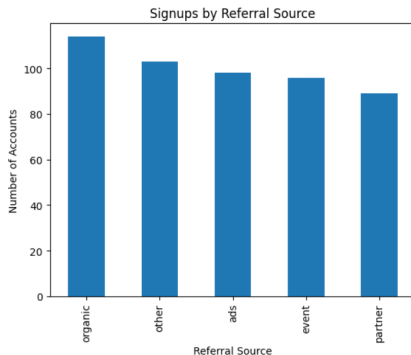
**Insight** The majority of customers operate in technology sectors, particularly software development and financial technology. These industries typically rely heavily on scalable software platforms and digital infrastructure.

# Customer Acquisition Channels

Customers are acquired through multiple referral sources.

| Referral Source | Customers |
|-----------------|-----------|
| Organic         | 114       |
| Other           | 103       |
| Ads             | 98        |
| Events          | 96        |
| Partner         | 89        |

# Graph and Insight



## Insight

Organic acquisition is the largest source of customers, suggesting strong brand visibility, search presence, or word-of-mouth referrals.

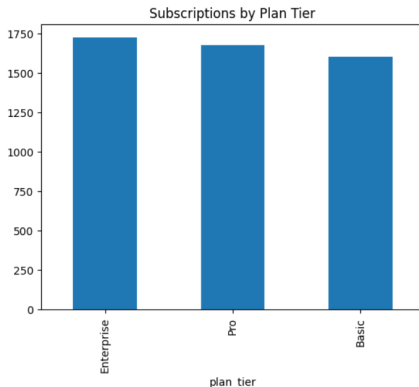
# Subscription Plan Distribution

Customers are distributed across three primary pricing tiers.

| Plan Tier  | Customers |
|------------|-----------|
| Enterprise | 1723      |
| Pro        | 1675      |
| Basic      | 1602      |



# Graph and Insight



## Insight

The relatively balanced distribution across pricing tiers suggests that the platform serves both smaller customers and larger enterprise clients.

# Customer Support Metrics

Customer support performance is an important factor in customer satisfaction.

## Key Support Metrics

- Average Ticket Resolution Time: 35 hours 51 minutes
- Average Customer Satisfaction Score: 3.98 / 5

## Ticket Priority Distribution

- Urgent: 514
- High: 510
- Medium: 491
- Low: 485

## Insight

Support demand is relatively evenly distributed across priority levels, suggesting a consistent volume of technical support requests.

# Most Used Features

| feature_name | usage_count |
|--------------|-------------|
| feature_32   | 6686        |
| feature_15   | 6621        |
| feature_6    | 6546        |
| feature_20   | 6541        |
| feature_34   | 6536        |
| feature_12   | 6534        |
| feature_11   | 6533        |
| feature_2    | 6525        |
| feature_38   | 6478        |
| feature_26   | 6470        |
| feature_17   | 6458        |
| feature_39   | 6430        |
| feature_31   | 6425        |
| feature_36   | 6389        |
| feature_24   | 6388        |
| feature_10   | 6384        |
| feature_37   | 6380        |
| feature_4    | 6374        |
| feature_22   | 6366        |
| feature_1    | 6364        |

# Feature Usage Analysis

The platform includes 40 major features used by customers.

## Most Frequently Used Features

- feature\_32 (6686 uses)
- feature\_15 (6621 uses)
- feature\_6 (6546 uses)
- feature\_20 (6541 uses)
- feature\_34 (6536 uses)

## Insight

High feature usage indicates that customers actively engage with the platform and rely on its capabilities for daily operations.

# Feature Errors

| error_count  |     |
|--------------|-----|
| feature_name |     |
| feature_4    | 418 |
| feature_26   | 417 |
| feature_9    | 404 |
| feature_2    | 401 |
| feature_34   | 397 |
| feature_16   | 396 |
| feature_40   | 380 |
| feature_13   | 372 |
| feature_29   | 371 |
| feature_36   | 371 |
| feature_18   | 369 |
| feature_17   | 369 |
| feature_27   | 362 |
| feature_30   | 362 |
| feature_28   | 361 |
| feature_12   | 359 |
| feature_32   | 357 |
| feature_39   | 357 |
| feature_38   | 356 |
| feature_19   | 352 |

# Feature Error Analysis

Certain features experience higher system error rates.

| Feature    | Error Rate (%) |
|------------|----------------|
| feature_4  | 6.56           |
| feature_9  | 6.51           |
| feature_26 | 6.45           |
| feature_16 | 6.31           |
| feature_18 | 6.31           |

## Insight

Higher error rates in frequently used features may negatively affect the user experience and contribute to customer dissatisfaction.

# Churn Rate Analysis

Out of 5,000 total subscriptions:

- 600 customers churned
- Observed churn rate: 12%

However, the overall churn rate across the customer lifecycle is approximately:

- 22%

## Interpretation

This means that more than one-fifth of customers eventually stop using the platform, indicating a significant opportunity for improving customer retention.

# Upgrade Rate Analysis

The platform shows an upgrade rate of:

- 10.58%

## Interpretation

Approximately one out of every ten customers upgrades to a higher subscription tier.

This indicates that some customers perceive increasing value in the platform as their usage grows.

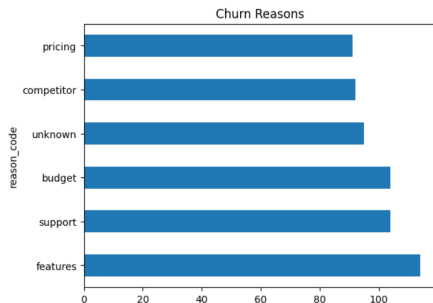
However, the churn rate is still significantly higher than the upgrade rate.



# Reasons for Customer Churn

| Reason     | Customers |
|------------|-----------|
| Features   | 114       |
| Support    | 104       |
| Budget     | 104       |
| Unknown    | 95        |
| Competitor | 92        |
| Pricing    | 91        |

# Graph and Insight



## Key Insight

The leading cause of churn is missing or inadequate features, followed by support issues and pricing concerns.

# Customer Usage vs Churn

Average usage comparison:

| Churn Status       | Average Usage |
|--------------------|---------------|
| Retained Customers | 50.23         |
| Churned Customers  | 51.13         |

## Insight

Customers who churn actually use the platform at similar levels as retained customers.

This suggests that churn is not caused by lack of engagement, but rather by dissatisfaction with features, pricing, or support.

The business generates strong recurring revenue.

## Financial Metrics

- Monthly Recurring Revenue (MRR): \$11,338,747
- Annual Recurring Revenue (ARR): \$136,064,964
- Customer Lifetime Value (CLV): \$10,307.95

## Insight

Because each customer generates significant lifetime value, reducing churn even slightly could produce substantial financial gains.

The analysis reveals several important findings:

- Customers actively use the platform features
- Feature limitations are the leading cause of churn
- Support quality influences customer satisfaction
- Pricing sensitivity affects some customers
- Error-prone features may negatively impact retention

Improving these areas could significantly reduce churn and increase customer lifetime value.

# Strategic Recommendations

To improve retention and revenue growth, the company should:

- Improve product features based on customer feedback
- Reduce system errors in high-usage features
- Enhance customer support responsiveness
- Introduce flexible pricing structures
- Improve onboarding and customer education

These improvements would strengthen customer relationships and increase long-term platform adoption.

# Conclusion

This data analytics project analyzed customer behavior, product usage, and support performance to understand churn patterns in a subscription-based business.

Key findings include:

- Churn rate of approximately 22%
- Upgrade rate of 10.58%
- Feature limitations and support issues are major churn drivers
- High customer lifetime value emphasizes the importance of retention

By addressing product quality, pricing, and support services, the company can reduce churn and improve long-term revenue growth.